Quadratic polynomials

Upper high school · Present and explore

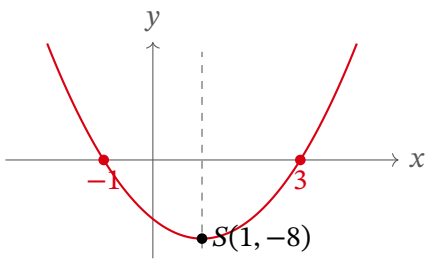
Definition · Expanded form. For $f(x) = ax^2 + bx + c$, with $a, b, c \in \mathbb{R}$ and $a \neq 0$, the domain is $\mathbb{R}$. The sign of a determines the opening of the parabola.

Property · Vertex form. $f(x) = a(x-\alpha)^2 + \beta$, where $\alpha = -\dfrac{b}{2a}$ and $\beta = f(\alpha) = c - \dfrac{b^2}{4a}$. The vertex is $S(\alpha, \beta)$ and the axis of symmetry is $x = \alpha$. If $a > 0$, f decreases on $(-\infty, \alpha]$ and increases on $[\alpha, +\infty)$; β is its minimum. If $a < 0$, the directions reverse and β is its maximum.

Property · Discriminant and roots. $\Delta = b^2 - 4ac$ and $\beta = -\dfrac{\Delta}{4a}$. If $\Delta > 0$, the two roots are $\dfrac{-b-\sqrt{\Delta}}{2a}$ and $\dfrac{-b+\sqrt{\Delta}}{2a}$: order them before naming them $x_1 < x_2$ (the order of these two formulas depends on the sign of a). If $\Delta = 0$, the double root is $x_0 = \alpha$. If $\Delta < 0$, there are no real roots.

Key point · Factorisation and sign. If $\Delta > 0$, $f(x) = a(x-x_1)(x-x_2)$. Outside the roots the two factors have the same sign; between them they have opposite signs. Thus the polynomial has the sign of a outside, the opposite sign between, and is zero at the roots. If $\Delta = 0$, $f(x) = a(x-x_0)^2$ has the sign of a except at x_0. If $\Delta < 0$, f has strictly the sign of a throughout $\mathbb{R}$ and cannot be factorised into real linear factors.

Example 1. $2x^2 - 4x - 6 = 2(x-1)^2 - 8 = 2(x+1)(x-3)$. The vertex is $(1, -8)$, $\Delta = 64$ and the roots are -1 and 3. Hence $f(x) \leq 0$ is equivalent to $x \in [-1, 3]$.



Activity 1 · Investigate and justify. Find the variation, roots and sign of $-x^2 + 2x + 3$, $x^2 - 2$ and $x^2 + x + 1$. Give exact roots, then approximations explicitly marked with $\approx$. Check your results by expanding a factorisation or substituting the roots.